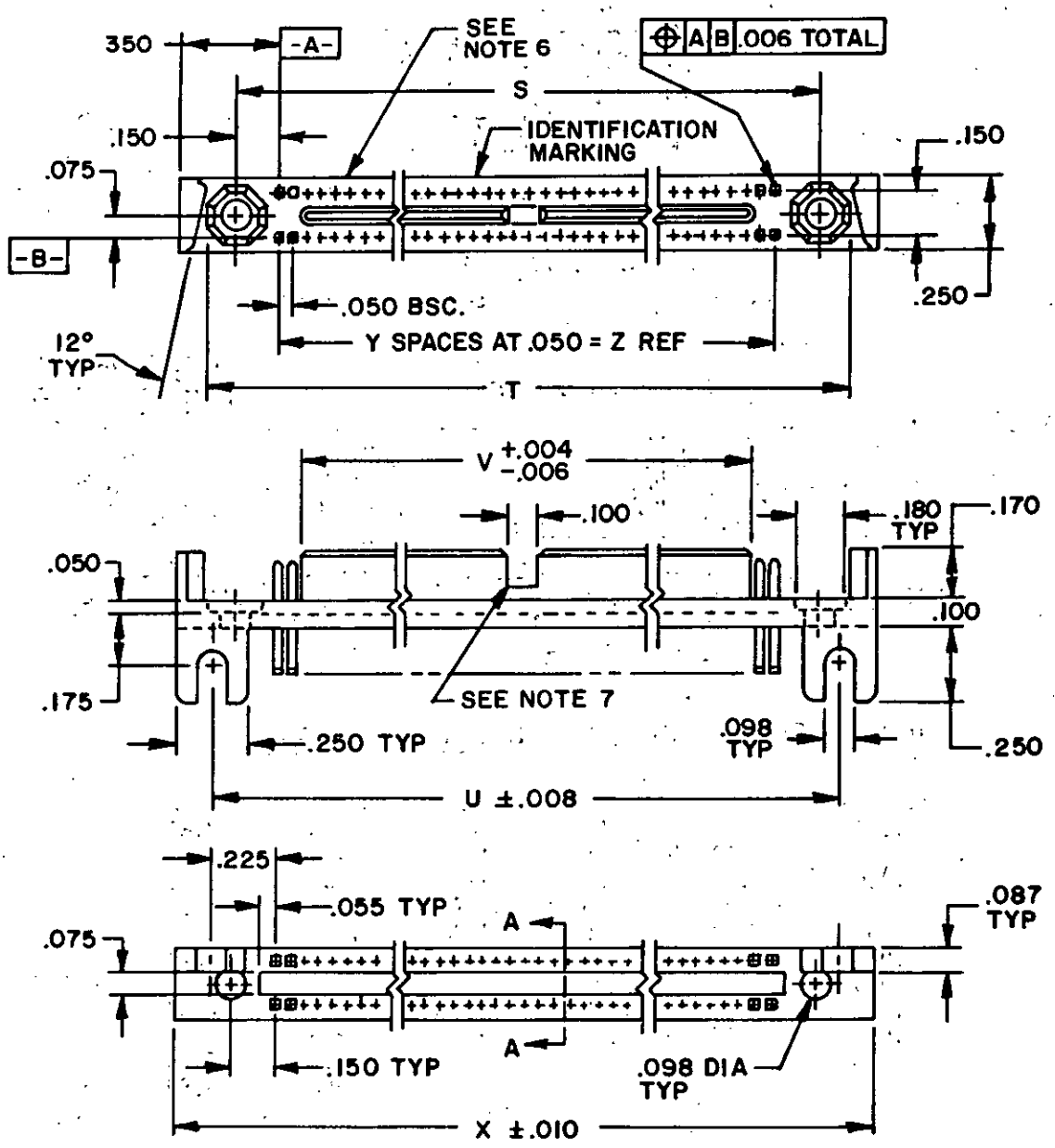
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, DECADE INCREMENT 20 THRU 110 AND 128 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.050 SPACING)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

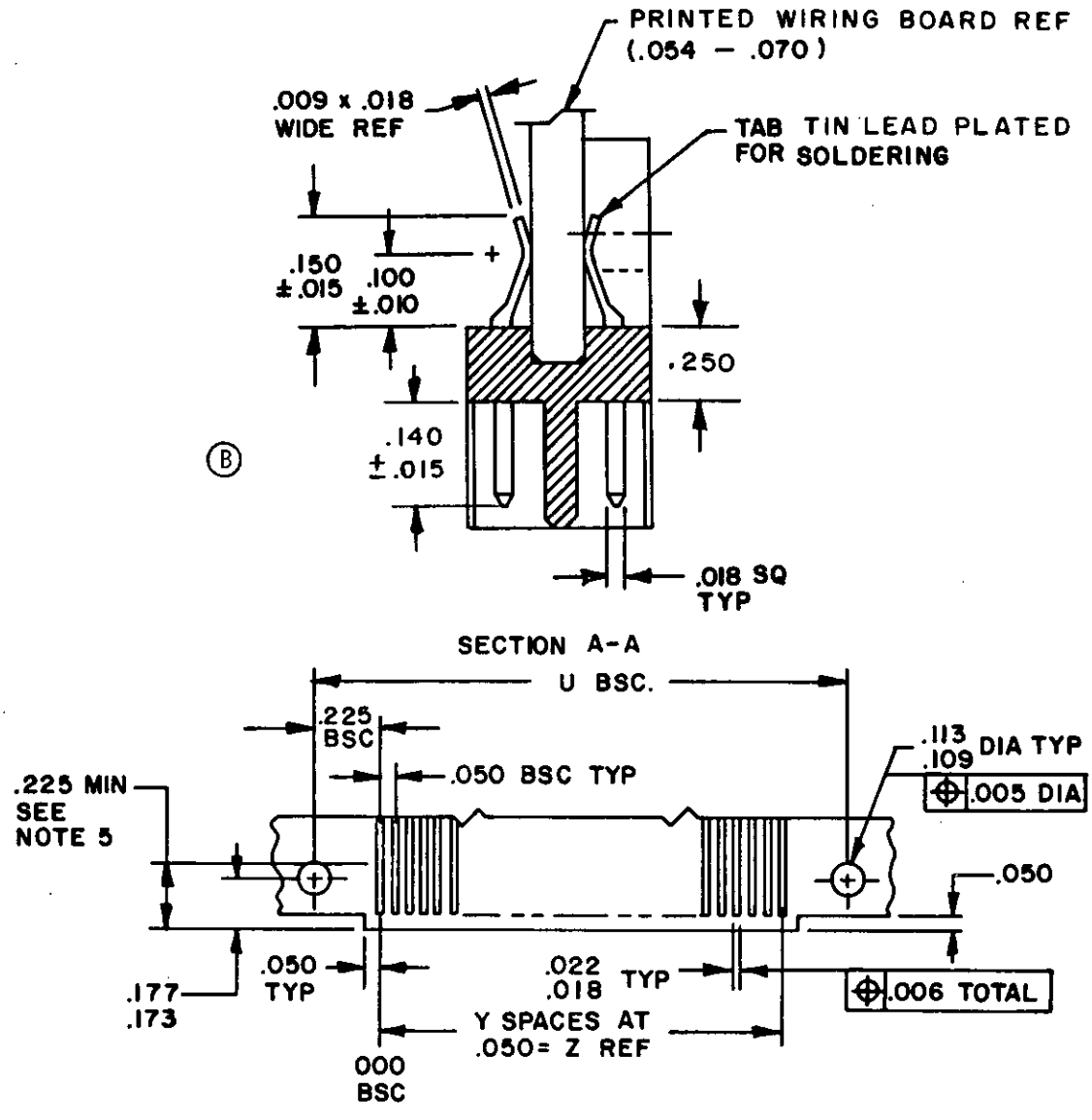


Ⓑ FIGURE 1. Connectors, plug without pin protectors .050 (1.27 mm) spacing for style 1.

Ⓑ denotes changes

FSC 5935

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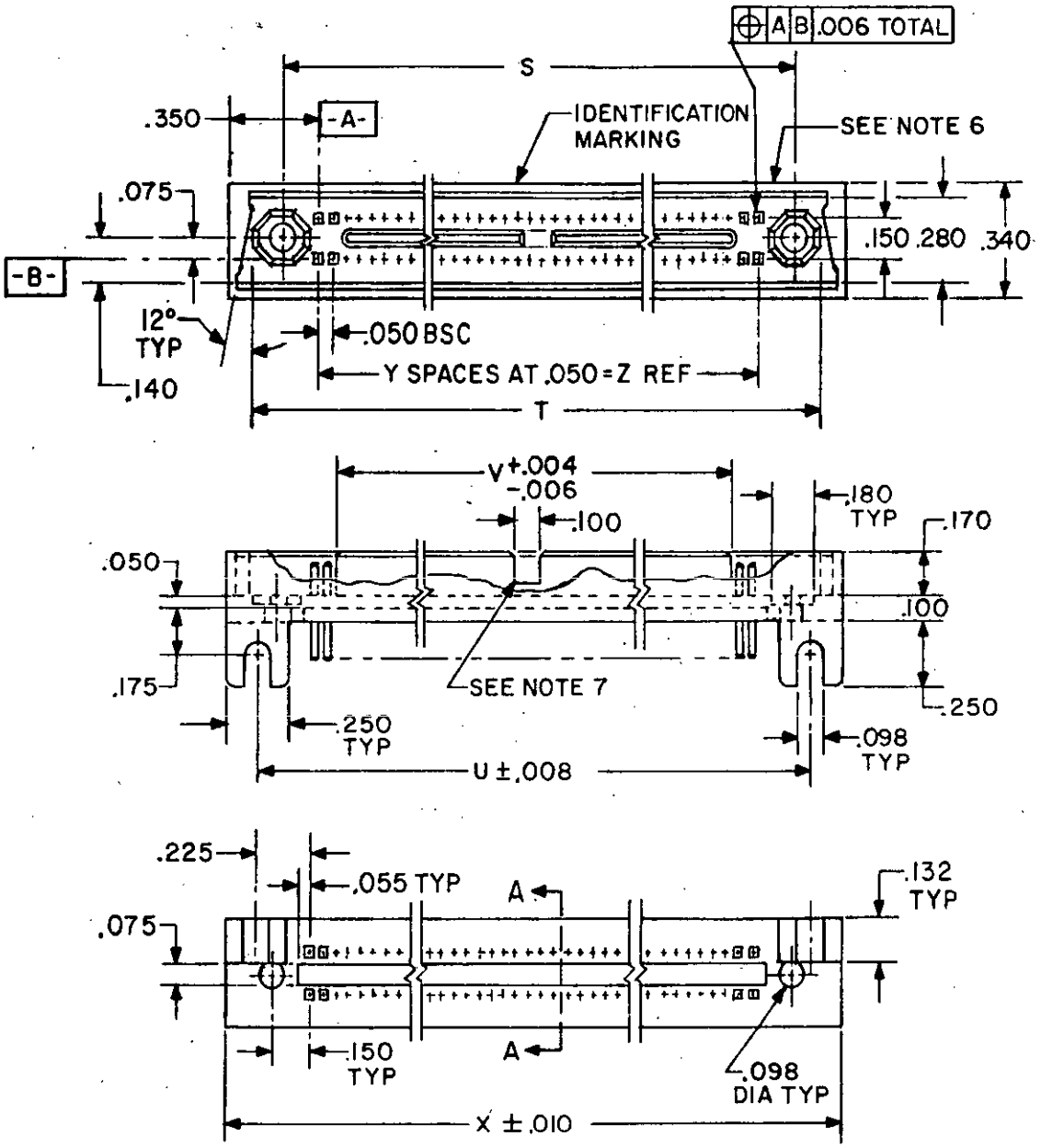


RECOMMENDED PRINTED WIRING BOARD LAYOUT

INCHES	MM	INCHES	MM	INCHES	MM
.004	.10	.050	1.27	.113	2.87
.005	.13	.054	1.37	.140	3.56
.006	.15	.055	1.40	.150	3.81
.008	.20	.070	1.78	.170	4.32
.009	.23	.075	1.90	.173	4.39
.010	.25	.087	2.21	.175	4.44
.015	.38	.098	2.49	.177	4.50
.018	.46	.100	2.54	.180	4.57
.022	.56	.109	2.77	.225	5.72
				.250	6.35
				.350	8.89

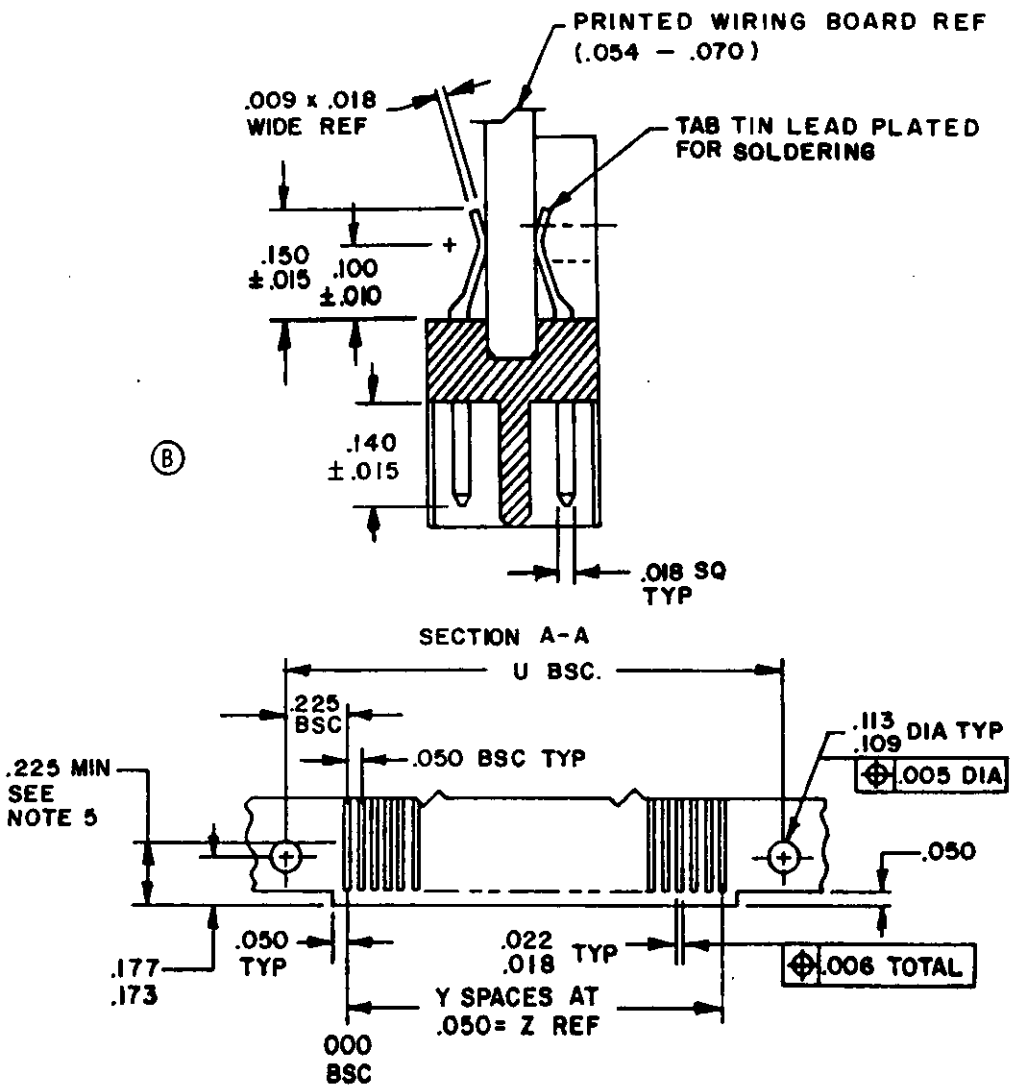
Ⓑ FIGURE 1. Connectors, plug without pin protectors .050 (1.27 mm) spacing for style 1 - Continued.

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⑧ FIGURE 2. Connectors, plug with pin protectors .050 (1.27 mm) spacing for style 2.

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RECOMMENDED PRINTED WIRING BOARD LAYOUT

INCHES	MM	INCHES	MM	INCHES	MM
.004	.10	.054	1.37	.150	3.81
.005	.13	.055	1.40	.170	4.32
.006	.15	.070	1.78	.173	4.39
.008	.20	.075	1.90	.175	4.44
.009	.23	.098	2.49	.177	4.50
.010	.25	.100	2.54	.180	4.57
.015	.38	.109	2.77	.225	5.71
.018	.46	.113	2.87	.250	6.35
.022	.56	.132	3.35	.280	7.11
.050	1.27	.140	3.56	.340	8.64
				.350	8.89

Ⓑ FIGURE 2. Connectors, plug with pin protectors .050 (1.27 mm) spacing for style 2 - Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/119(USAF).
(B) 5. Minimum pad length on printed wiring board.
6. Numbers indicating end positions stamped this side only.
7. Except 20 thru 50 position.

TABLE I. Dash numbers and dimensions. (Style 1)

Dash number	No. of contacts	Dimensions							Contact identification numbers
		S	T	U	V	X	Y	Z	
01	20	.750 (19.05)	.970 (24.64)	.900 (22.86)	.290 (7.37)	1.150 (29.21)	9	.450 (11.43)	1-10
02	30	1.000 (25.40)	1.220 (30.99)	1.150 (29.21)	.540 (13.72)	1.400 (35.56)	14	.700 (17.78)	1-15
03	40	1.250 (31.75)	1.470 (37.34)	1.400 (35.56)	.790 (20.07)	1.650 (41.91)	19	.950 (24.13)	1-20
04	50	1.500 (38.10)	1.720 (43.69)	1.650 (41.91)	1.040 (26.42)	1.900 (48.26)	24	1.200 (30.48)	1-25
05	60	1.750 (44.45)	1.970 (50.04)	1.900 (48.26)	1.290 (32.77)	2.150 (54.61)	29	1.450 (36.83)	1-30
06	70	2.000 (50.80)	2.220 (56.39)	2.150 (54.61)	1.540 (39.12)	2.400 (60.96)	34	1.700 (43.18)	1-35
07	80	2.250 (57.15)	2.470 (62.74)	2.400 (60.96)	1.790 (45.47)	2.650 (67.31)	39	1.950 (49.53)	1-40
08	90	2.500 (63.50)	2.720 (69.09)	2.650 (67.31)	2.040 (51.82)	2.900 (73.66)	44	2.200 (55.88)	1-45
09	100	2.750 (69.85)	2.970 (75.44)	2.900 (73.66)	2.290 (58.17)	3.150 (80.01)	49	2.450 (62.23)	1-50
10	110	3.000 (76.20)	3.220 (81.79)	3.150 (80.01)	2.540 (64.52)	3.400 (86.36)	54	2.700 (68.58)	1-55
11	128	3.450 (87.63)	3.670 (93.22)	3.600 (91.44)	2.990 (75.95)	3.850 (97.79)	63	3.150 (80.01)	1-64

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Ⓑ TABLE II. Dash numbers and dimensions. (Style 2)

Dash number	No. of contacts	Dimensions							Contact identification numbers
		S	T	U	V	X	Y	Z	
12	20	.750 (19.05)	.970 (24.64)	.900 (22.86)	.290 (7.37)	1.150 (29.21)	9	.450 (11.43)	1-10
13	30	1.000 (25.40)	1.220 (30.99)	1.150 (29.21)	.540 (13.72)	1.400 (35.56)	14	.700 (17.78)	1-15
14	40	1.250 (31.75)	1.470 (37.34)	1.400 (35.56)	.790 (20.07)	1.650 (41.91)	19	.950 (24.13)	1-20
15	50	1.500 (38.10)	1.720 (43.69)	1.650 (41.91)	1.040 (26.42)	1.900 (48.26)	24	1.200 (30.48)	1-25
16	60	1.750 (44.45)	1.970 (50.04)	1.900 (48.26)	1.290 (32.77)	2.150 (54.61)	29	1.450 (36.83)	1-30
17	70	2.000 (50.80)	2.220 (56.39)	2.150 (54.61)	1.540 (39.12)	2.400 (60.96)	34	1.700 (43.18)	1-35
18	80	2.250 (57.15)	2.470 (62.74)	2.400 (60.96)	1.790 (45.47)	2.650 (67.31)	39	1.950 (49.53)	1-40
19	90	2.500 (63.50)	2.720 (69.09)	2.650 (67.31)	2.040 (51.82)	2.900 (73.66)	44	2.200 (55.88)	1-45
20	100	2.750 (69.85)	2.970 (75.44)	2.900 (73.66)	2.290 (58.17)	3.150 (80.01)	49	2.450 (62.23)	1-50
21	110	3.000 (76.20)	3.220 (81.79)	3.150 (80.01)	2.540 (64.52)	3.400 (86.36)	54	2.700 (68.58)	1-55
22	128	3.450 (87.63)	3.670 (93.22)	3.600 (91.44)	2.990 (75.95)	3.850 (97.79)	63	3.150 (80.01)	1-64

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REQUIREMENTS:

Design and construction:

(B) Dimensions and configuration: See figures 1 and 2.

Material:

Body: Type GPT-30F per MIL-M-24519.

(B) Plating (contact): Contact plating shall be gold in accordance with MIL-G-45204, type II, grade C, class I over nickel in accordance with QQ-N-290, class 2, 30 to 150 microinches overall. If selective plating process is used, contact plating shall be gold in accordance with MIL-G-45204, type II, grade C, class I over nickel in accordance with QQ-N-290, class 2, 30 to 150 microinches on contact engagement area for .100 inch minimum length. The termination area plating shall be tin lead per MIL-T-10727, class I, .0001 minimum, the tin lead shall be 50 to 90 percent tin over nickel per QQ-N-290, class 2, 30 to 150 microinches, for .165 inch minimum length.

(B) Contact identification: See figures 1 and 2.

Contact rating: 1.5 amperes per contact, maximum.

Operating temperature: -65°C to 125°C.

Keying: See MIL-C-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .30 and the withdrawal force in pounds shall be a minimum of .03 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .015 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: .75 pound, minimum.

Dielectric withstanding voltage:

Sea level: 600 volts rms.

High altitude: 150 volts rms.

Solderability: Each connector shall be subjected to method 208 of MIL-STD-202 (solder termination only).

(B) Part number: M55302/118- (dash number from tables I and II).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
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a. Paragraph Number and Wording:

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